

Mock Midterm Exam 1

(6:35 PM – 7:50 PM : 75 Minutes)

- This exam will account for 20% of your overall grade.
- There are five (5) questions, worth 75 points in total. Please answer all of them.
- This is a *closed book, closed notes* exam. *No cheat sheets* are allowed.
- You are allowed to *use scratch papers* for your calculations.

GOOD LUCK!

Question	Parts	Points
1. Asymptotic Bounds	$(a), (b), (c)$	$5 + 5 + 5 = 15$
2. Recurrences.	$(a), (b), (c)$	$5 + 5 + 5 = 15$
3. Cutting Rod	–	12
4. Insertion Sort	–	18
5. Prefix Sums	$(a), (b)$	$10 + 5 = 15$
Total		75

QUESTION 1. [15 Points] Asymptotic Bounds. For each of the following statements state whether it is *True* or *False*. You must justify each answer. Assume that all log's are of base 2.

(a) [5 Points] $2^{\log n^2} = \Theta(n^2)$

(b) [5 Points] $n^{1.2} = \Omega(n \log^2 n)$

(c) [5 Points] $2^{2n} = \mathcal{O}(3^n)$

QUESTION 2. [15 Points] Recurrences. For each of the functions below either solve it using the Master Theorem or state why the Master Theorem does not apply.

$$(a) \text{ [5 Points] } \frac{1}{4}T(n) = \begin{cases} \Theta(1), & \text{if } n \leq 1; \\ 4T\left(\frac{n}{4}\right) + n^2, & \text{otherwise.} \end{cases}$$

$$(b) \text{ [5 Points] } T(n) = \begin{cases} \Theta(1), & \text{if } n \leq 1; \\ 3T\left(\frac{n}{2}\right) + (-1)^n n^2, & \text{otherwise.} \end{cases}$$

$$(c) \text{ [5 Points] } T(n) = \begin{cases} \Theta(1), & \text{if } n \leq 1; \\ 5T\left(\frac{n}{3}\right) + \Theta(n), & \text{otherwise.} \end{cases}$$

QUESTION 3. [12 Points] Cutting Rod. What the following algorithm EXTENDED-CUT-ROD($p[1..n]$, n) does is not important, but if you must know it does the following. Given a rod of length n inches and a list of prices $p[1..n]$, where $p[i]$ is the selling price of a rod of length i inches, it determines the maximum revenue $r[n]$ obtainable by cutting up the rod and selling the pieces.

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EXTENDED-CUT-ROD (  $p[1..n]$ ,  $n$  )
1.   $r[0..n] \leftarrow$  new array,  $s[0..n] \leftarrow$  new array
2.   $r[0] \leftarrow 0$ 
3.  for  $j \leftarrow 1$  to  $n$  do
4.     $q \leftarrow -\infty$ 
5.    for  $i \leftarrow 1$  to  $j$  do
6.      if  $q < p[i] + r[j - i]$  then
7.         $q \leftarrow p[i] + r[j - i]$ 
8.         $s[j] \leftarrow i$ 
9.     $r[j] \leftarrow q$ 
10. return  $r$  and  $s$ 

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Analyze the worst-case running time of EXTENDED-CUT-ROD on an input of size n .

QUESTION 4. [18 Points] Insertion Sort. Given an array $A[1..n]$ of n distinct numbers as input the *Insertion Sort* algorithm rearranges the items of A in nondecreasing order of value from left (smaller array index) to right (larger array index). Recall that the outermost **for** loop of the algorithm iterates from $j = 2$ to $j = A.length$.

Suppose the following array $A[1..10]$ is given as input to the algorithm. Please show the state of this array after each iteration of the outermost **for** loop.

	1	2	3	4	5	6	7	8	9	10
A	12	8	5	10	15	2	9	18	13	4

There must be one line for each value of j . Please start each line with the j value followed by a “.” and then write down the items (separated by commas, no fancy boxes are needed) of the input array in the order they will appear in $A[1..10]$ at the end of the j -th iteration of the loop.

QUESTION 5. [15 Points] Prefix Sums. The PREFIX-SUM algorithm given below accepts an array $A[p..r]$ of real numbers and updates $S[p : r]$ with prefix sums, that is, for $p \leq i \leq r$, sets $S[i] = \sum_{j=p}^i A[j]$. In order to compute the prefix sums of array $A[1..n]$, PREFIX-SUMS is called with $p = 1$ and $r = n$.

PREFIX-SUM (A, p, r, S)

1. **if** $p = r$
2. $S[p] = A[p]$
3. **else**
4. // split $A[p..r]$ into two approximately equal halves $A[p..q]$ and $A[q + 1..r]$
5. $q = \left\lfloor \frac{p+r}{2} \right\rfloor$
6. // recursively find prefix sums of the left half
7. **PREFIX-SUM** (A, p, q, S)
8. // recursively find prefix sums of the right half
9. **PREFIX-SUM** (A, q + 1, r, S)
10. // update prefix sums of right half with the total sum of the left half
11. **for** $i = q + 1$ **to** r
12. $S[i] = S[i] + S[q]$

- (a) [10 Points] Write a recurrence relation describing the running time of PREFIX-SUM on an array of length n . Assume for simplicity that $n = 2^k$ for some integer $k \geq 0$.
- (b) [5 Points] Solve the recurrence relation from part (a).